

Technology Stack

Date	29 June 2026
Team ID	
Project Name	A comprehensive measure of well being
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	React.js with Chart.js	Component reusability and a responsive, interactive prediction dashboard.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python (FastAPI)	Fast integration with the scikit-learn ML pipeline and strong existing team Python expertise.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	PostgreSQL	Strong relational structure fits tabular UNDP indicator data and supports SQL-based analytics.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	AWS (EC2 + S3) with Docker	Easy scalability and cost-effective hosting for both demo and production.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	GitHub + GitHub Actions	Seamless team collaboration and automated testing/deployment pipeline.
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	UNDP Human Development Report API, World Bank Open Data API	Provides authoritative, up-to-date life expectancy, education, and income indicators.

